

Mechanochemical Pd-Catalyzed Cross-Coupling of Arylhalides and Organozinc Pivalates

Tomáš Čarný,^[a] Tibor Peňaška,^[a] Samuel Andrejčák,^[a] and Radovan Šebesta^{*[a]}

Dedicated to Prof. Dr. Štefan Toma on the occasion of his 85th birthday.

Abstract: Cross-coupling reactions are essential tools in target molecule synthesis. However, the use of highly reactive organometallic reagents limits their applicability. Here, we present a mechanochemical Pd-catalyzed cross-coupling reaction between aryl halides and organozinc pivalates that can be carried out under ambient temperature and atmos-

phere. This operationally simple procedure affords a wide range of biaryl and aryl-heteroaryl derivatives in high yields and short times. The reaction tolerates various functional groups and can be realized on a synthetically useful scale. Its practical value was demonstrated in the short synthesis of the pharmaceutical diflunisal.

Introduction

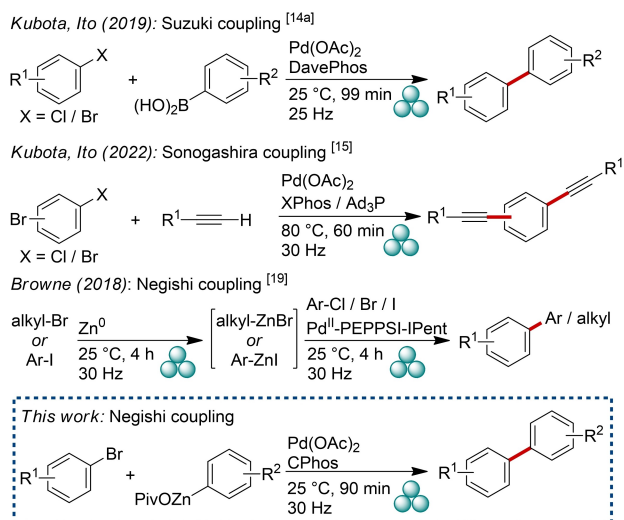
Transition-metal-catalyzed cross-coupling reactions significantly expanded the synthetic repertoire of organic chemistry and broadened our ability to provide compounds for medicine, materials, or crop protection.^[1] Although many cross-coupling reactions have become an almost routine part of the research, challenges remain. The inferior stability of transition-metal catalysts and the need for complex ligands limit the practicality of cross-coupling methods. Therefore, user-friendly and benign conditions, as well as operational simplicity, are highly desirable but largely undeveloped for cross-coupling reactions. Towards this goal, mechanochemistry appears to be a highly promising approach. Mechanochemically induced transformations can offer benefits such as short reaction times, high yields, small amounts of undesirable side-products, no or limited amount of solvent used, and operational simplicity.^[2] Furthermore, mechanochemical reactions may also produce products or selectivities that cannot be accessed via conventional means.^[3] Even though conceptual understanding of mechanochemistry is still under intense investigation,^[4] ball-milling is increasingly applied for the synthesis of important bioactive compounds,^[2f,5] or materials.^[6] Metal-catalyzed or mediated reactions can also benefit from mechanochemical activation.^[7] C–H bond activations using Pd, Rh, Ir, or Co-catalysis were effectively realized under mechanochemical ball-milling conditions.^[8] Reactions involving organometallic reagents or catalysts typically require rigorous control of reaction parameters and often require strictly anhydrous and oxygen-free conditions. These require-

ments seemingly contradict mechanochemical reactions that are typically done under normal conditions in the air. Interestingly, mechanochemistry was able to afford various air and moisture-sensitive compounds, hinting at intriguing possibilities that may be available for cross-coupling reactions under ball-milling conditions.^[9] Suzuki cross-coupling reaction was found to be compatible with mechanochemical conditions although early attempts suffered from various limitations.^[10] Interesting approach here can be a direct mechanocatalysis,^[11] which allowed Borchardt to realize Suzuki reactions with Pd-balls as catalysts.^[12] The key advancement of the mechanochemical cross-coupling methodology was the stabilization of Pd-intermediates with diene ligands performed by Ito and Kubota. They could even obtain oxidative addition Pd-complexes by ball milling.^[13] Several Suzuki cross-couplings were then realized by this team which led to various aryl compounds.^[14] Toolbox of mechanochemical cross-couplings was recently expanded also to Sonogashira reaction, which afforded a range of polycyclic arenes (Scheme 1).^[15] Buchwald-Hartwig aminations were also achieved under mechanochemical conditions.^[16]

Mechanochemical cross-coupling reactions with more reactive organometallics such as Grignard or organozinc reagents pose a bigger challenge due to the abovementioned sensitivity. Very recently, Kumada coupling was realized with Grignard reagents formed under ball-milling conditions.^[17] Organozinc reagents would be very useful as they are compatible with a wide range of functional groups. At the same time, organozinc reagents undergo smooth transmetalation with many catalytically relevant transition metals and thus are ideal candidates as nucleophilic partners in cross-coupling chemistry.^[18] Typical organozinc compounds are highly sensitive to moisture and oxygen, making their utilization difficult. Browne and coworkers developed a mechanochemical procedure for zinc activation with subsequent formation of typical organozinc bromides. These reagents were then utilized in mechanochemical Negishi cross-coupling reaction. However, the sensitivity of organozinc halides to air and moisture still hindered the practicality of this

[a] Dr. T. Čarný, Dr. T. Peňaška, S. Andrejčák, Prof. Dr. R. Šebesta
Department of Organic Chemistry, Faculty of Natural Sciences
Comenius University in Bratislava
Mlynská dolina, Ilkovičova 6, 842 15 Bratislava (Slovakia)
E-mail: radovan.sebesta@uniba.sk
Homepage: www.radovansebesta.com

Supporting information for this article is available on the WWW under <https://doi.org/10.1002/chem.202202040>



Scheme 1. Key mechanochemical cross-couplings and concept of this work.

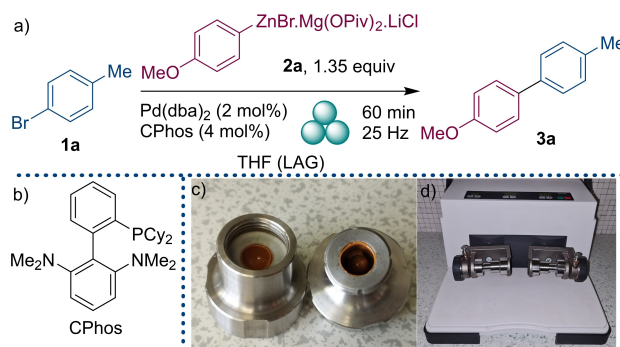
method and probably also lowered the yields of products.^[19] Another drawback of this otherwise very compelling method was the use of an expensive and sensitive Pd-PEPPSI catalyst (Scheme 1). Knochel developed a solid version of a modified organozinc reagent with pivalate anion, which can be conveniently prepared and handled on air.^[20] Moreover, these reagents are suitable partners in Pd, and Co-catalyzed cross-coupling reactions performed in solution.^[21] We hypothesized that such air stable and solid reagents would be ideal candidates for development of mechanochemical Negishi C(sp²)–C(sp²) cross-coupling.

In this context, we present Negishi cross-coupling of aryl halides with organozinc pivalates at ambient conditions using ball-milling.

Results and Discussion

We have started the exploration of the possible mechanochemical Negishi cross-coupling with 4-bromotoluene (**1a**) and 4-methoxyphenylzinc pivalate (**2a**)^[22] (Scheme 2). Among many possible conditions for cross-coupling reaction of organozinc reagents, we selected Pd(dba)₂ as catalyst and CPhos as a phosphine ligand.

All mechanochemical reactions were performed in a mixer mill in which milling jars make radial oscillations with frequencies from 3 to 30 Hz. The experiments were performed in stainless steel milling jars with an internal volume of 1.5 mL and with stainless steel balls (ø 6 mm). Initial reaction conditions with THF as LAG (liquid-assisted grinding) solvent afforded 60% conversion and 21% isolated yield of product **3a** (Table 1, entry 1). Several other bulky phosphine ligands such as cataCXium A, RuPhos, BrettPhos, DavePhos, or tBuPhos did not proceed with significant improvement. Increasing the catalyst loading to 10 mol% of Pd₂(dba)₃ and 20 mol% of CPhos ligand as well as longer reaction time (180 min) and frequency (30 Hz)



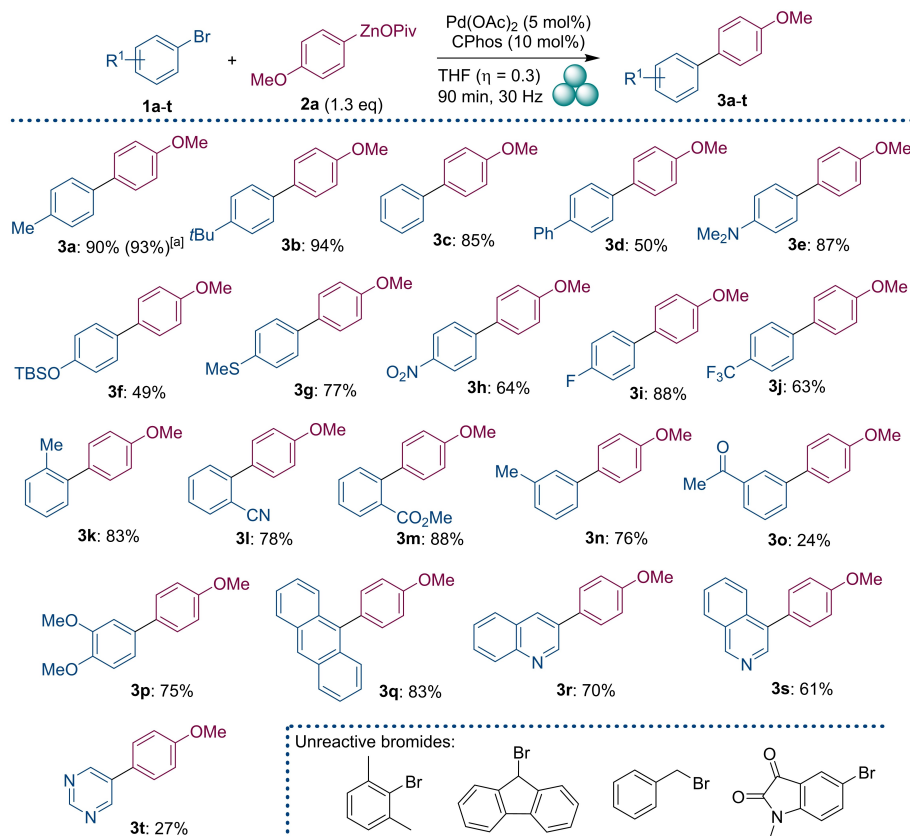
Scheme 2. a) Initial set-up for mechanochemical Negishi cross-coupling with organozinc pivalates; b) Structure of the most efficient ligand CPhos; c) milling jars with reaction mixture after completion of the reaction; d) mixer mill used for all experiments.

Table 1. Optimization of reaction conditions for the mechanochemical Negishi cross-coupling of bromide **1a** and zinc pivalate **2a**.^[a]

Entry	Pd-catalyst ([mol %])	Ligand ([mol %])	Time [min] frequency [Hz]	Conv. [%] ^[b]	NMR, ^[c] (Isolated) yield [%]
1	Pd(dba) ₂ (2)	CPhos (4)	60 25	60	24 (21)
2	Pd(dba) ₂ (2)	DavePhos (4)	60 25	32	21
3	Pd(dba) ₂ (2)	BrettPhos (4)	60 25	36	10
4	Pd(dba) ₂ (10)	CPhos (20)	180 25	95	95 (83)
5	Pd(PPh ₃) ₄ (10)	CPhos (20)	180 30	25	5
6	Pd(MeCN) ₂ Cl ₂ (10)	CPhos (20)	180 30	100	100
7	Pd(OAc) ₂ (10)	–	180 30	70	70
8	Pd(OAc) ₂ (10)	CPhos (20)	180 30	100	100
9	Pd(OAc) ₂ (5)	CPhos (10)	60 30	100	100 (90)

[a] Reaction set-up. THF as LAG. [b] Conversion of **1a**. [c] Determined by ¹H NMR spectroscopy by using 1,3,5-trimethoxybenzene as an internal standard.

led to 95% NMR yield (83% isolated) (Table 1, entry 4; see Table S1–S3 for more details). Subsequently, several solvents (DMSO, acetone, acetonitrile, DCM, DMF, Me–THF) of variable polarities were screened as LAG additives. However, none of the tested solvents performed better than THF (see Table S6 for more details). Interestingly, typical cross-coupling precatalyst Pd(PPh₃)₄ was completely inefficient under our reaction conditions. Additional improvement was observed with change to Pd^{II} catalysts. Both Pd(MeCN)₂Cl₂ and Pd(OAc)₂ were highly efficient catalysts affording quantitative NMR yield of product **3a** alongside with complete conversion (Table 1, entries 5–6). Such reactivity allowed to decrease the loading of the catalyst to 5 mol% of Pd(OAc)₂ and 10 mol% of CPhos ligand as well as shortening the reaction time to 60 min. Under these conditions, the reaction still proceeded with complete conversion, affording



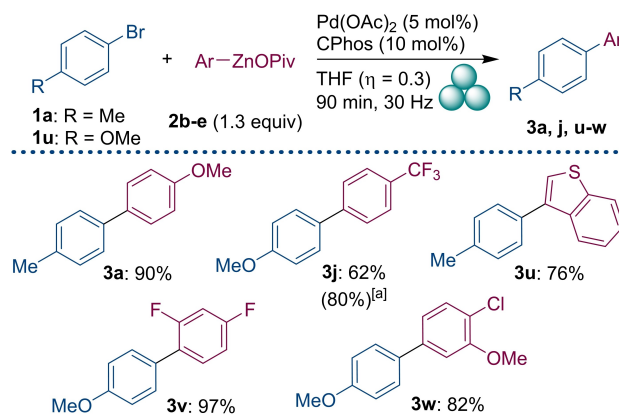
Scheme 3. Scope of the reaction with respect to aryl bromide; [a] Result from 1 mmol experiment.

product **3a** in a quantitative NMR yield (Table 1, entry 9). Under optimized conditions, the cross-coupling went to full conversion also with the corresponding triflate.

On the other hand, the cross-coupling reaction with 4-chlorotoluene and 4-iodotoluene were less efficient and afforded the coupling products in 22 and 66% yield (Table S9).

Due to their greater availability and lower costs, we have continued evaluating the scope of the reaction with aryl bromides. Under optimized reaction conditions and using zinc pivalate **2a**, we have screened a broad range of aryl and heteroaryl bromides. The mechanochemical Negishi cross-coupling afforded biaryl derivatives **3a–t** in high yields (Scheme 3). Aryl bromides with *ortho*, *meta*, and *para* substituent were all very well tolerated in this reaction. Both electron-donating as well as electron-withdrawing functional groups such as amino, methoxy, silyloxy, methylsulfanyl, nitro, fluoro, trifluoromethyl, nitril, ester, quinolinyll were compatible with this procedure. Somewhat lower yields were obtained for acetyl or pyrimidyl substituted starting materials. The reaction was unsuccessful for highly sterically hindered aryl bromide or aliphatic benzylbromide or with 5-bromo-1-methylindoline-2,3-dione (Scheme 3).

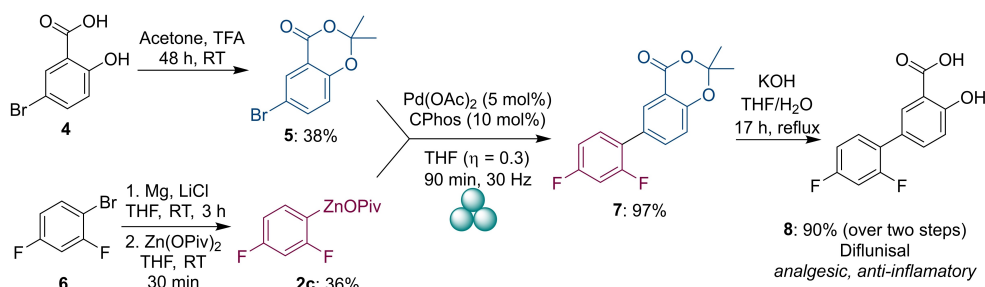
With 4-bromotoluene (**1a**) and 4-methoxybromobenzene (**1u**), we have evaluated several organozinc pivalates **2b–e** (Scheme 4). Various functionalities such as trifluoromethyl, chloro, fluoro, methoxy, or benzothiophenyl were well tolerated



Scheme 4. The reaction scope with respect to organozinc pivalates; [a] Result from 1 mmol experiment.

as organozinc coupling partner in the mechanochemical Negishi reaction. The corresponding biaryl products **3a**, **3j**, and **3u–w** were isolated in yields ranging from 62–97%. Interestingly, product **3j** was synthesized also by the other combination of aryl bromide and aryl pivalate reagents in a similar yield (cf. Schemes 3 and 4).

We have carried out also larger-scale experiments using milling jars with an internal volume of 10 mL and a ball with ϕ 10 mm. Starting with 1 mmol of bromide **1a** under optimized

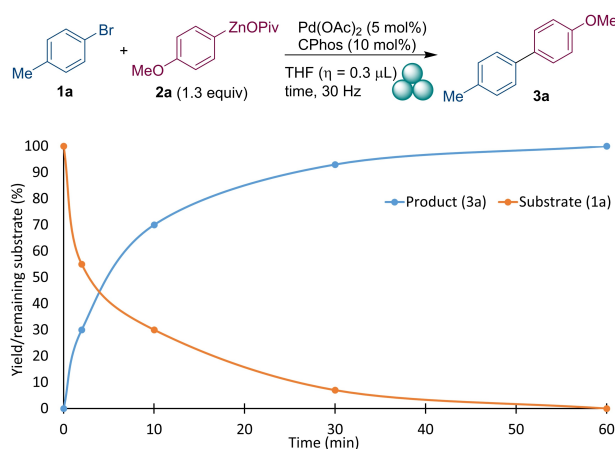


Scheme 5. Application of mechanochemical cross-coupling of organozinc pivalates in the synthesis of diflunisal.

conditions, product **3a** was obtained in 93% yield (185 mg). Similarly, 1 mmol reaction between bromide **1u** and pivalate **2b** furnished 80% of product **3j**.

To demonstrate the usefulness of this methodology, we used mechanochemical Negishi cross-coupling as a key step in the synthesis of analgetic and anti-inflammatory drug diflunisal (**10**).^[23] Several syntheses for diflunisal exist, and they typically start with 2,4-difluoroaniline and gradually build the target structure via several steps. Synthesis based on Negishi cross-coupling in the solution suggested that our mechanochemical strategy might also be successful.^[24] Our synthesis started from commercially available 5-bromosalicylic acid (**4**) and 2,4-difluorobromobenzene (**6**). Hydroxyl and carboxylic functions were protected as acetonide **5** and corresponding organozinc pivalate **2c** was generated from bromide **6** under standard conditions. Mechanochemical Negishi cross-coupling under optimized conditions provided biaryl **7** in almost quantitative yield. This material was directly subjected to basic deprotection, which furnished diflunisal (**8**) in 90% yield over two steps (Scheme 5).

We have observed the progress of the reaction between bromide **1a** and pivalate **2a** in time by ¹H NMR. The formation of the product **3a** corresponds to the consumption of the starting material (**1a**). Already after 30 min, there is more than 90% of product **3a** (Scheme 6).



Scheme 6. Reaction progress followed by ¹H NMR spectroscopy by using 1,3,5-trimethoxybenzene as an internal standard; Each data point was obtained from an individual reaction.

To evaluate the reduction of the amount of the used solvent, we have calculated mass intensity (MI). Mass intensity is defined as the total mass of reagents and solvents used in a reaction process relative to the mass of the isolated product (does not account for the purification of the reaction).^[25] Ball milling cross-coupling of bromide **1a** with pivalate **2a** under otherwise standard conditions has MI of 18.2. On the other hand, the corresponding reaction in the solution (with the minimum amount of THF needed to dissolve reagents) has a considerably higher MI of 82.7 (see Supporting Information for details of the calculation).

Conclusion

In conclusion, we have developed a mechanochemical Negishi cross-coupling reaction of organozinc pivalates with aryl bromides. This palladium-catalyzed transformation uses easily prepared air-stable organozinc reagents. Simple Pd(OAc)₂ as a Pd-source and bulky monophosphine ligand CPhos with a minimum amount of THF as LAG additive and 60–90 min grinding afforded a broad range of biaryl products in high yields. The transformation is also applicable on a synthetically useful scale. The methodology also improved the synthesis of analgetic medicine diflunisal. Applying liquid-assisted grinding to solid organozinc pivalates allowed to perform Negishi cross-coupling reaction under operationally simple conditions thus significantly improving practical utility of this key cross-coupling transformation.

Experimental Section

General procedure for mechanochemical Negishi cross-coupling

A stainless steel milling jar (internal volume 1.5 mL) was charged with corresponding arylhalide (0.146 mmol), Pd(OAc)₂ (2.0 mg, 0.007 mmol, 5 mol%), CPhos (6.0 mg, 0.015 mmol, 10 mol%), organozinc pivalate (1.3 equiv, its concentration was determined by titration with iodine) and anhydrous THF (LAG, η = 0.3 μL/mg). Milling jar equipped with a steel ball (diameter 6 mm) was closed, reaction mixture was milled from 60 to 90 min at 30 Hz at ambient temperature. Milling jar was opened, reaction mixture was dissolved in DCM, filtrated through a pad of celite and the filtrate was concentrated under a reduced pressure. The crude mixture was

purified by column chromatography to afford the desired coupling product.

Acknowledgements

This work was supported by the Slovak Grant Agency VEGA, grant no. VEGA 1/0332/19.

Conflict of Interest

The authors declare no conflict of interest.

Data Availability Statement

The data that support the findings of this study are available in the supplementary material of this article.

Keywords: biaryl · cross-coupling · liquid-assisted grinding · mechanochemistry · organozinc

- [1] A. de Meijere, S. Bräse, M. Oestreich, *Metal Catalyzed Cross-Coupling Reactions and More*, Wiley-VCH, Weinheim, 2013.
- [2] a) T. Friščić, C. Mottillo, H. M. Titi, *Angew. Chem. Int. Ed.* **2020**, *59*, 1018–1029; *Angew. Chem.* **2020**, *132*, 1030–1041; b) J. Howard, Q. Cao, D. L. Browne, *Chem. Sci.* **2018**, *9*, 3080–3094; c) J.-L. Do, T. Friščić, *ACS Cent. Sci.* **2017**, *3*, 13–19; d) D. Tan, T. Friščić, *Eur. J. Org. Chem.* **2018**, *2018*, 18–33; e) J.-L. Do, T. Friščić, *Synlett* **2017**, *28*, 2066–2092; f) D. Tan, L. Loots, T. Friscic, *Chem. Commun.* **2016**, *52*, 7760–7781; g) G. A. Bowmaker, *Chem. Commun.* **2013**, *49*, 334–348; h) S. L. James, C. J. Adams, C. Bolm, D. Braga, P. Collier, T. Friscic, F. Grepioni, K. D. M. Harris, G. Hyett, W. Jones, A. Krebs, J. Mack, L. Maini, A. G. Orpen, I. P. Parkin, W. C. Shearouse, J. W. Steed, D. C. Waddell, *Chem. Soc. Rev.* **2012**, *41*, 413–447; i) A. Stolle, T. Szuppa, S. E. S. Leonhardt, B. Ondruschka, *Chem. Soc. Rev.* **2011**, *40*, 2317–2329; j) B. Rodríguez, A. Bruckmann, T. Rantanen, C. Bolm, *Adv. Synth. Catal.* **2007**, *349*, 2213–2233.
- [3] J. G. Hernández, C. Bolm, *J. Org. Chem.* **2017**, *82*, 4007–4019.
- [4] a) R. T. O'Neill, R. Boulatov, *Nat. Chem. Rev.* **2021**, *5*, 148–167; b) S. Lukin, L. S. Germann, T. Friščić, I. Halasz, *Acc. Chem. Res.* **2022**, *55*, 1262–1277; c) S. Mateti, M. Mathesh, Z. Liu, T. Tao, T. Ramireddy, A. M. Glushenkov, W. Yang, Y. I. Chen, *Chem. Commun.* **2021**, *57*, 1080–1092.
- [5] a) O. Bento, F. Luttringer, T. M. El Dine, N. Pétry, X. Bantreil, F. Lamaty, *Eur. J. Org. Chem.* **2022**, e202101516; b) P. Ying, J. Yu, W. Su, *Adv. Synth. Catal.* **2021**, *363*, 1246–1271.
- [6] D. Tan, F. García, *Chem. Soc. Rev.* **2019**, *48*, 2274–2292.
- [7] a) A. Porcheddu, E. Colacino, L. De Luca, F. Delogu, *ACS Catal.* **2020**, *10*, 8344–8394; b) J. G. Hernández, T. Friščić, *Tetrahedron Lett.* **2015**, *56*, 4253–4265.
- [8] J. G. Hernández, *Chem. Eur. J.* **2017**, *23*, 17157–17165.
- [9] K. Kubota, H. Ito, *Trends Chem.* **2020**, *2*, 1066–1081.
- [10] a) S. F. Nielsen, D. Peters, O. Axelsson, *Synth. Commun.* **2000**, *30*, 3501–3509; b) L. M. Klingensmith, N. E. Leadbeater, *Tetrahedron Lett.* **2003**, *44*, 765–768; c) F. Schneider, B. Ondruschka, *ChemSusChem* **2008**, *1*, 622–625; d) F. Schneider, A. Stolle, B. Ondruschka, H. Hopf, *Org. Process Res. Dev.* **2009**, *13*, 44–48; e) F. Bernhardt, R. Trotzki, T. Szuppa, A. Stolle, B. Ondruschka, *Beilstein J. Org. Chem.* **2010**, *6*, 7.
- [11] W. Pickhardt, S. Grätz, L. Borchardt, *Chem. Eur. J.* **2020**, *26*, 12903–12911.
- [12] a) W. Pickhardt, C. Beaković, M. Mayer, M. Wohlgemuth, F. J. Leon Kraus, M. Etter, S. Grätz, L. Borchardt, *Angew. Chem. Int. Ed.* **2022**, e202205003; b) C. G. Vogt, S. Grätz, S. Lukin, I. Halasz, M. Etter, J. D. Evans, L. Borchardt, *Angew. Chem. Int. Ed.* **2019**, *58*, 18942–18947; *Angew. Chem.* **2019**, *131*, 19118–19123.
- [13] K. Kubota, R. Takahashi, H. Ito, *Chem. Sci.* **2019**, *10*, 5837–5842.
- [14] a) T. Seo, T. Ishiyama, K. Kubota, H. Ito, *Chem. Sci.* **2019**, *10*, 8202–8210; b) T. Seo, K. Kubota, H. Ito, *J. Am. Chem. Soc.* **2020**, *142*, 9884–9889; c) T. Seo, N. Toyoshima, K. Kubota, H. Ito, *J. Am. Chem. Soc.* **2021**, *143*, 6165–6175; d) R. Takahashi, T. Seo, K. Kubota, H. Ito, *ACS Catal.* **2021**, *14803–14810*; e) K. Kubota, K. Kondo, T. Seo, H. Ito, *Synlett* **2022**, *33*, 898–902.
- [15] Y. Gao, C. Feng, T. Seo, K. Kubota, H. Ito, *Chem. Sci.* **2022**, *13*, 430–438.
- [16] a) Q.-L. Shao, Z.-J. Jiang, W.-K. Su, *Tetrahedron Lett.* **2018**, *59*, 2277–2280; b) Q. Cao, W. I. Nicholson, A. C. Jones, D. L. Browne, *Org. Biomol. Chem.* **2019**, *17*, 1722–1726; c) K. Kubota, T. Seo, K. Koide, Y. Hasegawa, H. Ito, *Nat. Commun.* **2019**, *10*, 111.
- [17] R. Takahashi, A. Hu, P. Gao, Y. Gao, Y. Pang, T. Seo, J. Jiang, S. Maeda, H. Takaya, K. Kubota, H. Ito, *Nat. Commun.* **2021**, *12*, 6691.
- [18] D. Haas, J. M. Hammann, R. Greiner, P. Knochel, *ACS Catal.* **2016**, *6*, 1540–1552.
- [19] Q. Cao, J. L. Howard, E. Wheatley, D. L. Browne, *Angew. Chem. Int. Ed.* **2018**, *57*, 11339–11343; *Angew. Chem.* **2018**, *130*, 11509–11513.
- [20] S. Bernhardt, G. Manolikakes, T. Kunz, P. Knochel, *Angew. Chem. Int. Ed.* **2011**, *50*, 9205–9209; *Angew. Chem.* **2011**, *123*, 9372–9376.
- [21] a) Y.-H. Chen, M. Ellwart, V. Malakhov, P. Knochel, *Synthesis* **2017**, *49*, 3215–3223; b) F. H. Lutter, S. Graßl, L. Grokenberger, M. S. Hofmayer, Y.-H. Chen, P. Knochel, *ChemCatChem* **2019**, *11*, 5188–5197.
- [22] M. Ellwart, Y.-H. Chen, C. P. Tüllmann, V. Malakhov, P. Knochel, *Org. Synth.* **2018**, *95*, 127–141.
- [23] J. Hannah, W. V. Ruyle, H. Jones, A. R. Matzuk, K. W. Kelly, B. E. Witzel, W. J. Holtz, R. A. Houser, T. Y. Shen, *J. Med. Chem.* **1978**, *21*, 1093–1100.
- [24] C. Jeong Nak, J. Lee Sang, H. Jo Gi, S. Jeong, *Bull. Korean Chem. Soc.* **2000**, *21*, 165–166.
- [25] D. J. C. Constable, A. D. Curzons, V. L. Cunningham, *Green Chem.* **2002**, *4*, 521–527.

Manuscript received: June 30, 2022
Accepted manuscript online: July 29, 2022
Version of record online: August 18, 2022